

Input: This is a knowledge graph completion task, which needs to predict the tail entity for an incomplete query triplet. The query triplet is (Valparaiso University, inverse relation of /location/location/contains, ?).

The quantized representation of entity Valparaiso University is [527] [1345] [1849] [1227] [1751] [2038] [818] [515] [1417] [333] [29] [721] [1691] [798] [1033] [153]

The answer candidates and corresponding quantized representations are as follows:

Minnesota, [1532] [258] [1837] [357] [923] [1994] [638] [555] [771] [1003] [1736] [1473] [1495] [1436] [1313] [20]
New York, [661] [1243] [542] [1741] [1907] [1799] [858] [1794] [1916] [458] [1844] [909] [438] [1737] [686] [963]
California, [1059] [1286] [1604] [846] [1086] [451] [1087] [1794] [994] [297] [1463] [159] [556] [1836] [407] [963]
Massachusetts, [202] [1243] [977] [757] [304] [389] [1172] [1308] [1916] [1858] [1323] [11] [841] [1680] [1798] [1885]
Illinois, [961] [1025] [1267] [174] [643] [1951] [1742] [1794] [1720] [1481] [543] [1883] [695] [1921] [182] [963]
New York City, [1458] [326] [1707] [239] [151] [640] [1366] [1794] [610] [458] [1844] [932] [122] [311] [121] [868]
United Kingdom, [51] [193] [1354] [669] [1867] [881] [480] [1271] [392] [1858] [650] [909] [1503] [1126] [1550] [153]
Pennsylvania, [361] [825] [1052] [1655] [1670] [732] [951] [1569] [275] [1995] [543] [4] [753] [351] [331] [637]
Los Angeles, [1584] [1231] [1707] [1461] [1867] [1466] [265] [1933] [850] [1533] [805] [1128] [1824] [1823] [307] [963]
Florida, [2016] [326] [542] [1614] [462] [1433] [1388] [819] [926] [1289] [1321] [563] [1977] [1144] [1268] [662]
Ohio, [1643] [1889] [1604] [88] [1364] [485] [1819] [1569] [54] [1582] [1500] [411] [438] [125] [1636] [20]
Texas, [2012] [1845] [1207] [412] [531] [1394] [1004] [688] [653] [1671] [1790] [1690] [1732] [1686] [1721] [1205]
Virginia, [99] [825] [738] [1859] [1287] [1540] [708] [780] [653] [662] [756] [1873] [1514] [1686] [59] [409]
England, [848] [1220] [1052] [590] [175] [451] [529] [1933] [808] [1598] [1790] [1988] [670] [524] [121] [136]
Connecticut, [633] [1025] [672] [338] [1694] [1799] [1528] [1177] [1949] [458] [1703] [411] [395] [33] [233] [1013]
Indiana, [497] [1875] [1849] [377] [1694] [61] [1471] [1445] [392] [1672] [1500] [300] [711] [1839] [331] [136]
Los Angeles County, [662] [1772] [558] [1623] [304] [1755] [1388] [1794] [1554] [333] [1662] [1258] [1474] [1764] [1021]
[409]
Louisiana, [1584] [326] [1309] [50] [1035] [0] [546] [728] [1130] [297] [1284] [11] [1514] [1929] [1248] [136]
Michigan, [172] [326] [382] [174] [1364] [1774] [1165] [664] [994] [1480] [1923] [431] [816] [1587] [1443] [860]
Hennepin County, [584] [1890] [1263] [1168] [1990] [656] [784] [1735] [102] [1533] [1979] [1005] [1312] [786] [873]
[1294]

Please generate quantized representations of the top-3 potential answer entities, ranked from highest to lowest:

LLM Output: 1, [497] [1875] [1849] [377] [1694] [61] [1471] [1445] [392] [1672] [1500] [300] [711] [1839] [331] [136]
2, [1532] [258] [1837] [357] [923] [1994] [638] [555] [771] [1003] [1736] [1473] [1495] [1436] [1313] [20]
3, [661] [1243] [542] [1741] [1907] [1799] [858] [1794] [1916] [458] [1844] [909] [438] [1737] [686] [963]
Ground Truth: [497] [1875] [1849] [377] [1694] [61] [1471] [1445] [392] [1672] [1500] [300] [711] [1839] [331] [136]